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SED Modeling of Circumstellar Disk Systems Using Dust Wall, Halo, and Disk Components

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ABSTRACT

Circumstellar disks provide insight into the structure and evolutionary stage of young stellar systems, especially when the systems cannot be spatially resolved. In this project, I modeled the spectral energy distributions of HD 169142, HD 32297, and HD 31293 using combinations of stellar blackbody emission, disk emission, dust walls, and halo components. Each model used a wavelength-dependent Planck function and a temperature-radius power law to represent dust emission from different disk geometries. The models were normalized to the stellar component and compared to observed λF_λ data in order to evaluate the relative strength of infrared excess emission. HD 32297 was best represented by a weak flat disk and small dust wall, producing an infrared-to-stellar luminosity ratio of $L_{\rm IR}/L_* = 0.009$, consistent with an evolved debris disk. HD 169142 required a stronger flared disk and inner dust wall. HD 31293 showed a similarly strong infrared excess which was best reproduced using a compact inner disk and extended dust halo. These results show that simple SED models can distinguish between weak debris disks and dust-rich young stellar environments, although degeneracies in normalization, dust temperature, and component geometry limit the physical precision of the derived parameters.

Keywords: Young Stellar Objects (1834) — Circumstellar disks (235) — Spectral energy distribution (2129)

1. INTRODUCTION

Circumstellar disks provide important clues about how young stellar systems evolve and how dust is distributed around protostars. Because many disk systems are difficult to resolve, spectral energy distribution modeling offers a way to infer disk structure from the amount of light emitted at different wavelengths. Infrared excess emission is especially useful because it traces dust that absorbs stellar radiation and re-emits it at longer wavelengths. I hypothesize that infrared excesses in these systems can be partly explained by additional dust structures beyond a simple stellar and disk blackbody model. Specifically, I test whether potential dust wall and halo components can account for overdensities of emission in the infrared and near each star's sublimation radius. In this project, I model three systems, HD 169142, HD 32297, and HD 31293, using combinations of stellar blackbody emission, disk emission, dust walls, and halo components to test which structures best reproduce their observed SEDs. All relevant code and object data can be found at https://github.com/mpmp23/phys-311-yso-modeling-final-project.

2. METHODS

While all wavelength data was measured in microns, the flux count of each dataset was initially in varying units – HD 32297 was F_ν in Janskys, HD 169142 was in λF_λ in CGS units, and HD 31293 was in νF_ν in CGS units, as well. All data was converted to λF_λ. As νF_ν and λF_λ are equivalent, only the Janskys data was converted into this λF_λ format directly.

Each dataset was given initial calculation parameters - first, temperatures were calculated using Wien's law using the first (inherently lowest) wavelength on the given dataset as its peak. This may not have been the protostar's technical peak, however the shortest-wavelength point is most likely to be dominated by direct stellar emission rather than cooler disk emission, making it a reasonable anchor for estimating the photospheric temperature of each young star.

Next, each system was given certain parameters to best model the relative data. All systems used the standard star and disk blackbody curves to model the normalized flux data. To gather this data, a starting and ending wavelength for each model based on the highest and lowest wavelength data limited the wavelength span of each SED. A wavelength step of 100 nm was used to calculate the flux at each given 100 nm increment in this range. At each increment, the peak temperature was used with that incremented wavelength to calculate the flux of the star. The same concept was applied to the disk – a sublimation radius R_{sub} and ending radius R_{end} for the disk controlled the range of the disk.

As opposed to the clear-cut stellar blackbody model, each model's disk would have a varying temperature as the radius of the disk increased. Similar to the wavelength structure, a step size of 0.05 AU for each model calculated the width of each small disk annulus. At these 0.05 AU increments, the area of each annulus $(R^2 - (R + 0.05)^2)$ and the midpoint temperature of each annulus was calculated in accordance with the Boltzmann-derived relationship $1800K(Rmid/Rsub)^{tpow}$, where 1800 K marked the sublimation temperature of the dust measured in each system. 1800 K was used as the consistent value across all models.

Two additional structures, a hypothetical dust wall and dust halo, were tested on each system and implemented if they helped to better map flux densities that could not be explained by the star and disk model alone. For the dust wall, I used the relationship $2R * height_w$ to determine area of the wall and $1800K * (R_{\text{wall}}/R_{\text{sublimation}})^{-1/2}$ to determine the temperature of the wall, assuming temperature uniformity across the wall. It should be noted that in the code itself, a wall height was given to model the area of this wall better, but during normalization, this area was factored out and did not dictate any part of the final flux determination. For the halo, I used an equivalent temperature estimation of $1800K * (R_{\text{halo}}/R_{\text{sublimation}})^{-1/2}$.

The underpinning of all models and structures consisted in Flux calculations using the Planck blackbody equation $B_\lambda(T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/(\lambda k_B T)}-1}$ which added up the varying intensity values given the temperature of each object from their peak wavelength (calculated for each star initially based on the lowest-wavelength light available on the data) and a wavelength from our pre-calculated wavelength arrays. Across all structures, flux was calculated using the aforementioned Planck function at each wavelength step, and for the disk, halo, and dust wall structures, each structure's area was multiplied by this to scale the strength of each model's intensity accordingly. Flux for each was summed to a F_{total} value, which was used for later normalization.

In order to understand the relative intensity of the emission in the disk, halo, and wall structures relative to the star post-normalization, a scaling factor was used to best map each model relative to their respective stars. Each structure was normalized by dividing by its summed flux so that the total integrated contribution of each component was 'unified' before applying a relative scaling factor.

Because the model SED was built from normalized blackbody components, its flux values were not automatically in the same units or magnitude as the measured object data. To correct for this, I chose a reference wavelength where the model and observed data should match and solved for a constant scaling factor:

$$LF_{\text{model}} = C\,(LF_{\text{data}})$$

That gives:

$$C = \frac{LF_{\text{model}}}{LF_{\text{data}}}$$

After calculating C, the entire observed dataset was multiplied by this value. This did not change the shape of the observed SED; it only shifted the data up or down so that it could be compared directly to the model. Each object received its own scaling factor because each system had a different flux scale and was anchored at a different wavelength point. For HD 169142 and HD 32297, the scaling was chosen near the near-infrared/stellar portion of the SED, while HD 31293 was scaled closer to its optical peak. This allowed the comparison to focus on whether the model reproduced the relative shape of the SED, especially the infrared excess caused by the disk, wall, or halo components.

From here, several values were calculated from this information.Calculating L_{IR}/L_* can provide meaningful information about the object's relative evolutionary stage. To estimate the relative infrared luminosity of each object, I integrated the modeled spectral energy distribution over wavelength. Because the models are plotted in terms of λF_λ, the relevant integral is taken over $d\ln\lambda$. This works because $d\ln\lambda = d\lambda/\lambda$, so $\lambda F_\lambda d\ln\lambda = F_\lambda d\lambda$. Therefore, the area under a λF_λ curve as a function of $\ln\lambda$ gives the total observed flux in that component.

The infrared-to-stellar luminosity ratio was then calculated by comparing the integrated flux from the infrared-emitting model components to the integrated flux from the stellar photosphere:

$$\frac{L_{\text{IR}}}{L_*} = \frac{\int \lambda F_{\lambda,\text{IR}}\, d\ln\lambda}{\int \lambda F_{\lambda,*}\, d\ln\lambda}.$$

For each object, $F_{\lambda,\mathrm{IR}}$ represents the sum of the non-stellar model components included in the fit. For HD 169142 and HD 32297, this includes the disk and dust wall components. For HD 31293, this includes the disk and halo components. Thus, the ratios were computed as:

$$\frac{L_{\mathrm{IR}}}{L_*} = \frac{\int \lambda F_{\lambda,\mathrm{disk}}\, d\ln\lambda + \int \lambda F_{\lambda,\mathrm{wall/halo}}\, d\ln\lambda}{\int \lambda F_{\lambda,*}\, d\ln\lambda}.$$

Although this calculation uses flux rather than absolute luminosity, the ratio is still valid because the stellar and infrared-emitting components are both part of the same system and therefore lie at the same distance (via the inverse square law). The factor needed to convert flux to luminosity, $4\pi d^2$, would appear in both the numerator and denominator and cancels out. Therefore, the integrated flux ratio provides an estimate of L_{IR}/L_*, instead of flux.

To estimate the approximate dust mass, I used this ratio, L_{IR}/L_*, to estimate the total luminosity emitted by the dust component. Assuming the dust radiates approximately like a blackbody at a characteristic dust temperature T_{dust}, the emitting surface area was then estimated using the Stefan-Boltzmann relation,

$$A_{\mathrm{emit}} = \frac{L_{\mathrm{dust}}}{\sigma T_{\mathrm{dust}}^4}.$$

From this emitting area, I made a rough estimate of the dust mass by assuming spherical grains with a representative grain radius and density. The resulting mass estimate was calculated as

$$M_{\mathrm{dust}} \approx \frac{A_{\mathrm{emit}} a \rho}{3},$$

where a is the assumed grain radius and ρ is the grain density. This method was only derived to give a relative approximation, since the true dust mass depends on grain size distribution, composition, opacity, and whether the dust emission is optically thin or optically thick.

3. RESULTS AND DISCUSSION

Overall, the results supported my hypothesis that additional dust structures were needed to explain the infrared excesses in these systems. The dust wall improved the fits for HD 169142 and HD 32297 near the inner disk/sublimation region, while the halo component better accounted for the strong extended infrared excess in HD 31293.

Table 1. Model Star Temperatures Based on Peak Wavelength

Object	**Star Temperature Based on Peak Wavelength (K)**
HD 169142	13,181.82
HD 32297	9,780.78
HD 31293	13,809.52

From the process outlined previously, a clear image of each system can be discerned; HD 169142 contains a flared disk far away from its star at 50 AU-200 AU, as well as a disk wall starting at 0.5 AU, the sublimation radius of this star. HD 32297 contains a flat disk starting at 18 AU away from its star, and ending at 70 AU, alongside a dust wall at the star's sublimation radius of 0.7 AU. HD 31293 reflects a system with an extremely small disk patch starting at the sublimation radius of 0.13 AU and ending at 0.5 AU, as well as a further-out halo of disk around the system hovering around 50 AU . As the range of the disk is so small, the scaling temperature-radius power law yielded almost identical results – therefore, labeling 31293's disk as 'flared' is likely a misnomer in our results, and is more reflective of an expanded dust wall around the star. Given what we know generally about each of these objects, these findings make sense. HD 32297 did indeed have two different dust zones, one wall near its sublimation radius and one much further away at 18 AU. HD 169142 had a large disk gap in between the contribution from the dust wall to the 50 AU starting radius of the dust. And, HD 31293's large IR excess can be attributed to a halo encompassing the system at a distant radius. In this project, the sublimation temperature of the dust pictured was assumed to be consistently 1800K, and the sublimation radii were determined not through a physics-derived calculation, but rather through a best-fit of the model's behavior. An important future question to consider would be to understand the characteristics

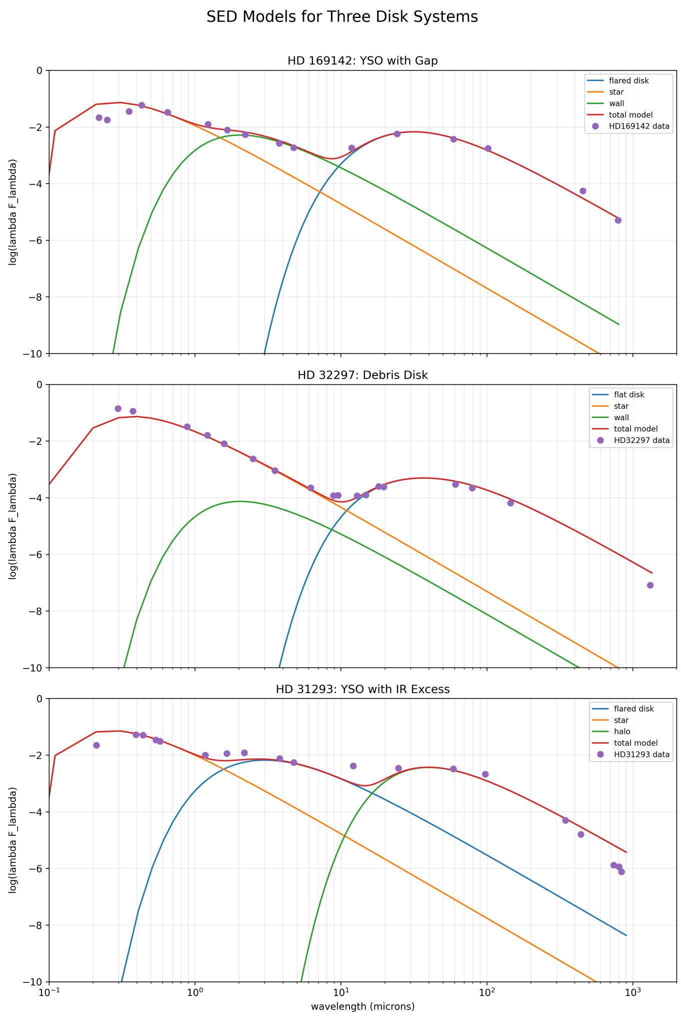

Figure 1. Model Results from Each System

Table 2. IR-to-Stellar Luminosity Ratios for Each Object

Object	$L_{\rm IR}/L_*$
HD 169142	0.17
HD 32297	0.009
HD 31293	0.15

of the kind of dust in the system, how this may impact sublimation temperature, as well as whether the sublimation radii identified would align with the estimated stellar temperatures of each system.

Calculating $L_{\rm IR}/L_{\rm star}$ appears to align with prior understanding of these system's relative developmental phase. Using parameters set by Wyatt (1) as precedent, HD 32297 corresponds with a debris disk (which was already known to be associated as a characteristic of 32297), which confirms the structure outlined in our graph and that our modeled flux count likely corresponds relatively well with this object's true nature.

These estimates of dust mass suggest that HD 169142 requires the largest emitting dust area and dust mass. HD 32297 has the smallest estimated dust mass, consistent with it being modeled as a weaker debris disk system. HD 31293 falls between the two, showing a larger dust contribution than HD 32297 but still less than HD 169142. Because these values depend strongly on the assumed stellar luminosities, dust temperatures, and grain properties, they should be interpreted as rough order-of-magnitude estimates rather than precise physical masses.

Table 3. Estimated Dust Emitting Areas and Dust Masses

Object	Emitting Area (m^2)	Dust Mass (kg)	Dust Mass (M_{Earth})
HD 169142	6.09×10^{24}	6.09×10^{21}	1.02×10^{-3}
HD 32297	1.42×10^{24}	1.42×10^{21}	2.38×10^{-4}
HD 31293	2.25×10^{24}	2.25×10^{21}	3.76×10^{-4}

Table 4. Relative Scale of Model Components to Their Star

Model Component	HD 32297	HD 169142	HD 31293
Wall Scale	0.1%	7%	–
Halo Scale	–	–	5%
Disk Scale	0.8%	10%	10%

The scaling factors pictured in Table 4 show the relative strength of each dust component compared to the normalized stellar emission, not the literal size or mass of the disk. HD 32297 has very small wall and disk scales, suggesting a weaker infrared excess consistent with a more evolved debris disk. HD 169142 and HD 31293 have much larger disk, wall, or halo scales, meaning their circumstellar dust contributes much more strongly to the total SED and is consistent with younger, dust-rich systems.

Despite attempts to fit the models as closely as possible to the given data, there are clear degeneracies in each graph; HD 169142 contains a close-fitting total model curve except for around the highest stellar peak. Degeneracies across all three objects had similar over or underestimations for the stellar peak. These could likely be due to a mis-calibration of spectral data. However, stars are not perfect blackbodies, either, so certain degeneracies could also be gas-blocked or under-fitted flux counts from the star. And, both HD 32297 and HD 31293 have over-estimations of the object's flux in the infrared, which likely means that their disks end closer than previously assumed, or the halo structure of HD 31293 is more sparse in infrared emission than previously assumed.

Outside of general model fit, I would likely adjust other important factors were I given sufficient time. Potentially a better normalization schema could avoid the elimination of the wall height in the dust wall example. More consideration could be put into a varying sublimation temperature – for example, dust at varying grain size or composition could influence at what precise temperature it could sublimate. 1800 K was a standard number, but given different consistencies and composition of dust grains, this number could be an overestimation. Outside of this research, an interesting direction could be to test what the relative opacity of such dust might be to get a better understanding of the look of these objects.

4. CONCLUSION

Overall, these models show that SED fitting can reveal broad differences in circumstellar structure, shape, and composition. The stronger infrared excesses of HD 169142 and HD 31293 suggest younger, dust-rich environments, while the weaker excess of HD 32297 is more consistent with an evolved debris disk with a dust wall. Using disk temperature-radius relationships alongside simple wall and halo components allowed the models to reproduce the major differences among the three systems. However, these results remain approximate because the fits depend on assumed sublimation temperatures, simplified geometries, and normalized component scaling rather than fully physical radiative transfer. Future work could improve the models by using independently measured stellar luminosities, dust opacities, grain-size distributions, and data-derived sublimation radii.

REFERENCES

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